

Global Renewable Energy in 2025

How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix, spanning hydropower, wind, and so

Source: Our World in Data — Renewable Energy

Authors: Hannah Ritchie, Max Roser, Pablo Rosado | Data: Ember (2026); Energy Institute (2025)

75% of Global GHG Emissions Still Come from Fossil Fuels

Fossil fuels have dominated energy systems since the Industrial Revolution, driving climate change and air pollution deaths

75% of GHG Emissions

Fossil fuels account for three-quarters of all greenhouse gas emissions globally.

5M+ Premature Deaths

Annual deaths attributed to fossil fuel air pollution worldwide.

Rapid Shift Required

The world must accelerate toward low-carbon energy — nuclear and renewables — to reduce CO₂ and improve air quality.

Why Renewables?

Renewable energy includes solar, wind, hydro, geothermal, bioenergy, wave, and tidal sources — clean, abundant, and rapidly scaling.

75%

of global GHG emissions

5M+

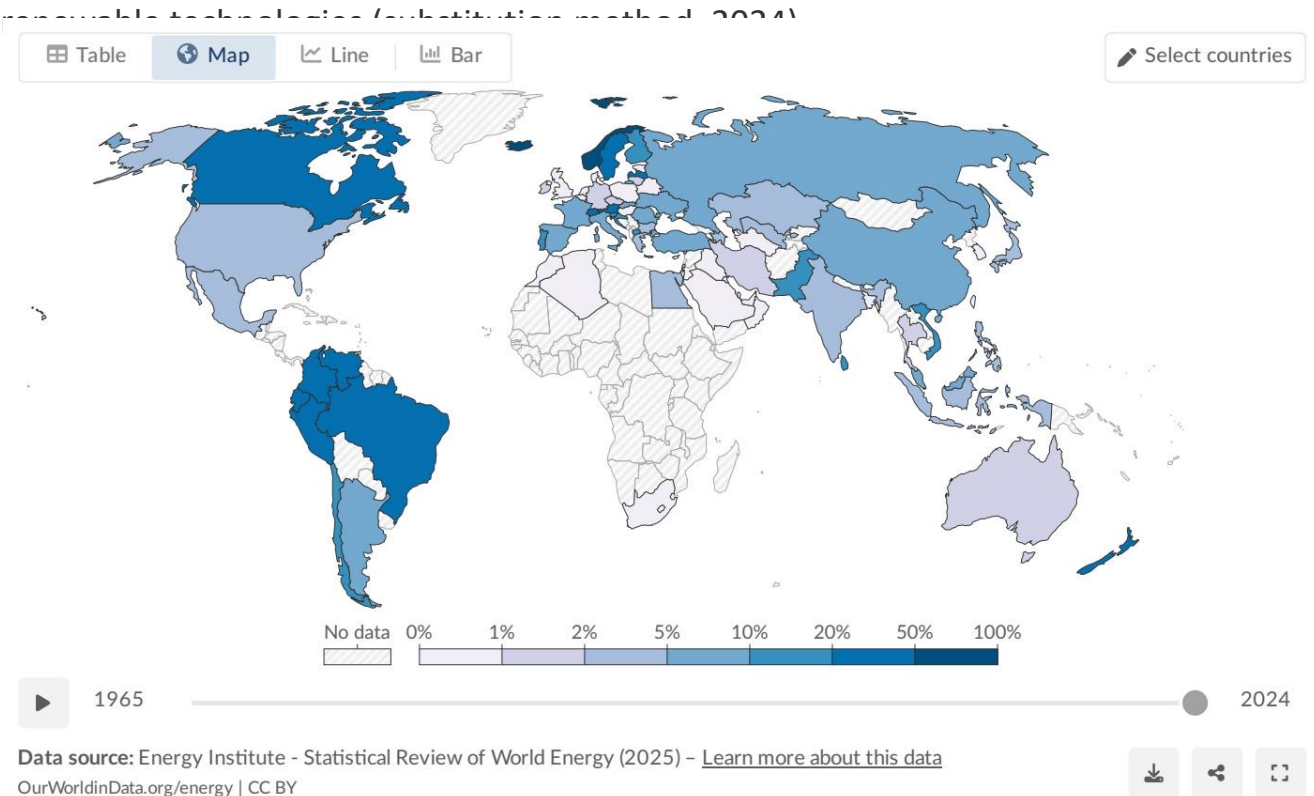
annual premature deaths

Renewables Now Supply ~14% of Global Primary Energy

Approximately one-seventh of the world's primary energy comes from renewable technologies (substitution method, 2024). Primary energy includes electricity, transport, and heating combined. Because transport and heating are harder to decarbonize (more reliant on oil and gas), the renewable share in the total energy mix is lower than in the electricity mix alone.

Key Leaders:

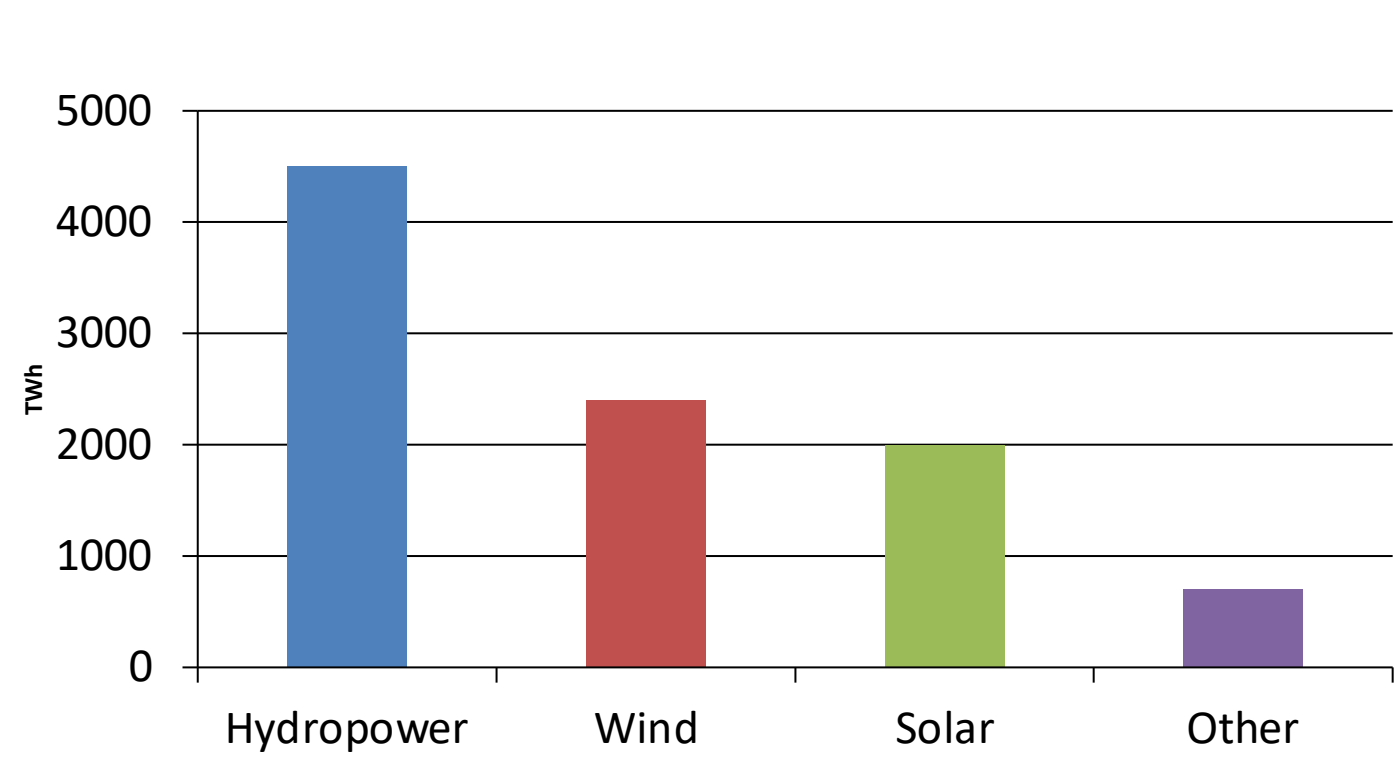
- Norway, Brazil, New Zealand — high renewable share
- Middle East & parts of Asia — below 5%
- Fossil-fuel-dependent nations face greatest transition challenge



Source: Our World in Data — Share of primary energy from renewables by country, 2024

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

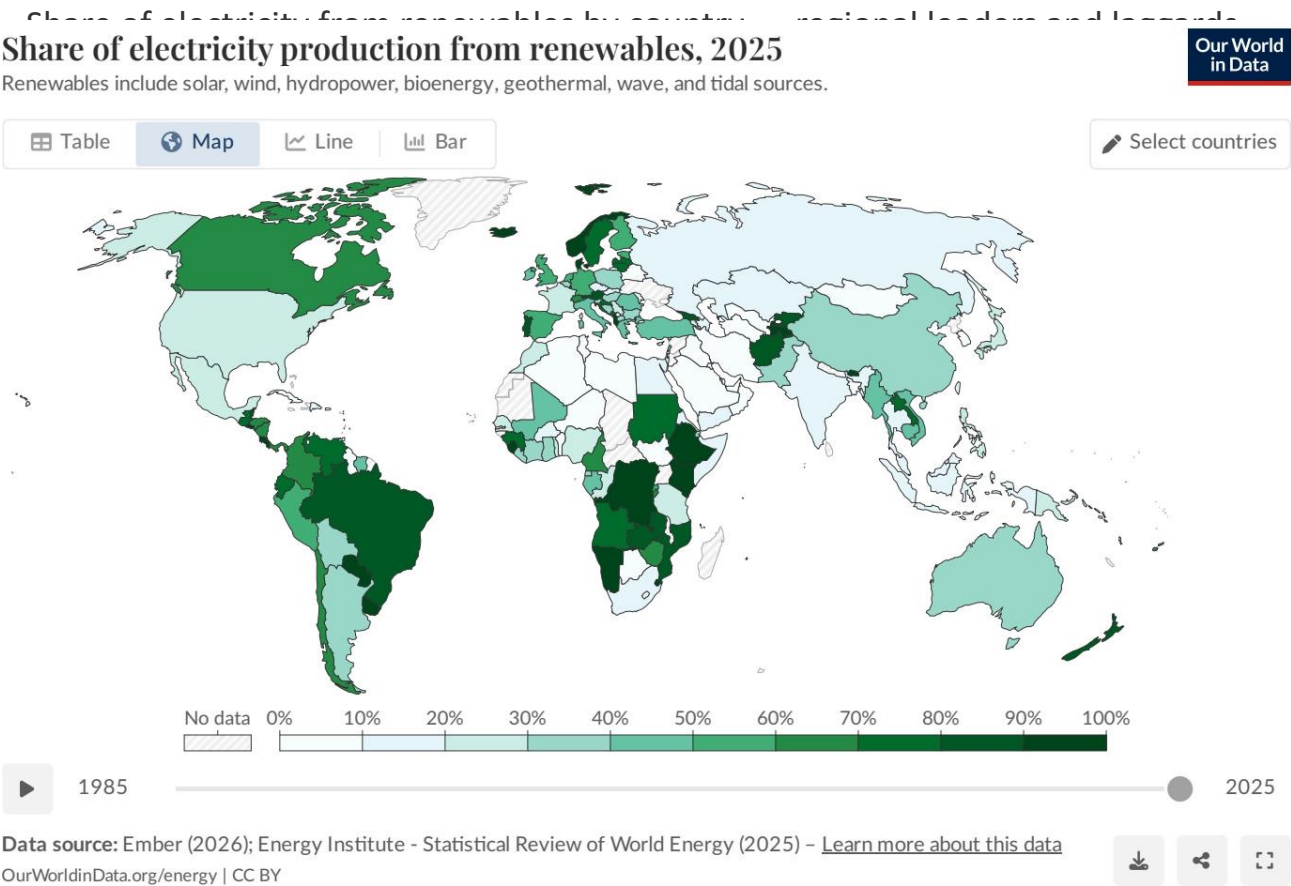
Global renewable electricity generation by technology — hydro, wind, solar, and other (TWh)



Technology	TWh	Share
Hydropower	~4,500	~47%
Wind	~2,400	~25%
Solar	~2,000	~21%
Other	~700	~7%

Wind and solar combined now nearly match hydropower's output — and are growing much faster.

Almost One-Third of Global Electricity Now Comes from Renewables



Country/Region	Renewable Share	Status
Norway	~98%	Leader
Brazil	~85%	Leader
Canada	~65%	Leader
Germany	~55%	Strong
China	~35%	Growing
India	~25%	Growing
World Average	~30%	—
Saudi Arabia	<2%	Lagging

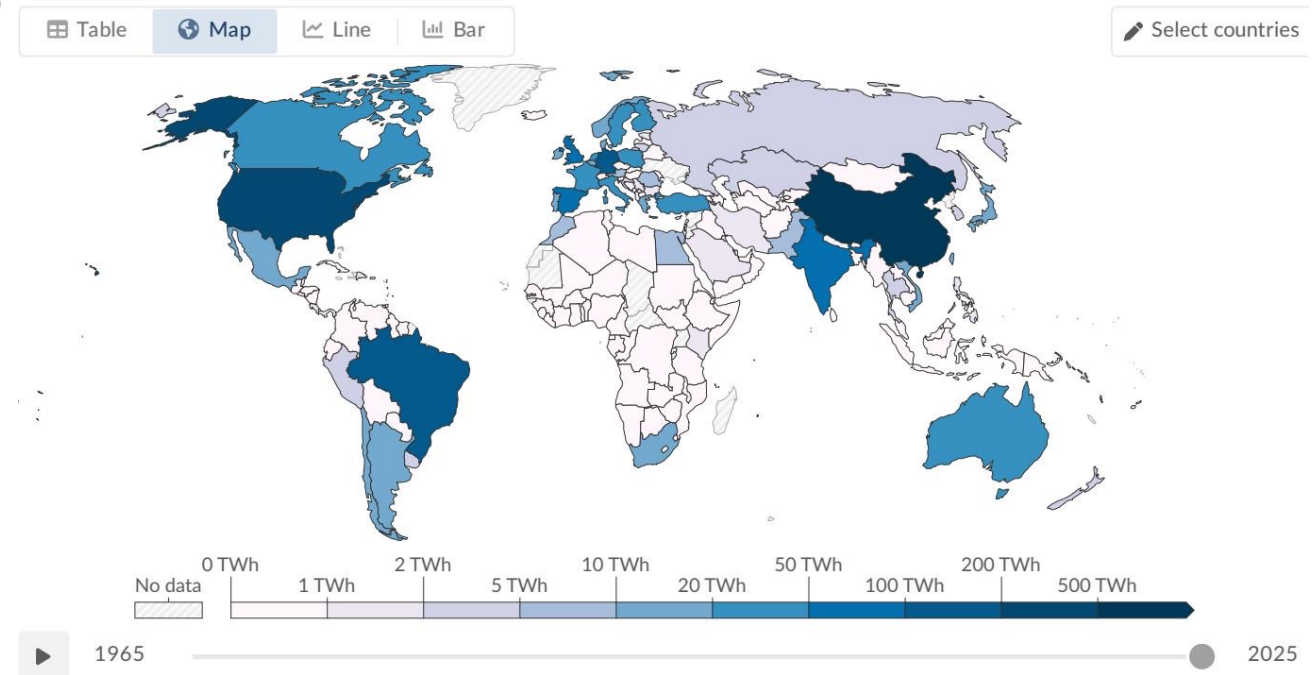
Source: Our World in Data — Share of electricity from renewables, 2025

Hydropower Remains the Largest Renewable Source at ~50% of Generation

Hydropower has generated electricity at scale for over a century — top producers: China, Brazil, Canada

Key Points:

- Hydropower accounts for roughly half of all renewable electricity generation worldwide.
- China, Brazil, and Canada are the top producers, each benefiting from vast water resources.
- Hydropower has been the backbone of renewable electricity for over 100 years.
- While dominant today, hydro's growth potential is more limited compared to wind and solar.
- Hydro plays a crucial role in grid stability and energy storage via pumped hydro.



~4,500 TWh generated | ~47% of all renewable electricity

Source: Our World in Data — Hydropower generation by country, 2025

Wind and Solar Are the Fastest-Growing Energy Technologies in History

Wind: ~2,400 TWh (~25%) | Solar: ~2,000 TWh (~21%) — combined, they nearly match hydropower

Wind Energy

~2,400 TWh

~25% of renewable electricity

Major growth in Europe, China, and the US

Onshore and offshore installations expanding

Solar Energy

~2,000 TWh

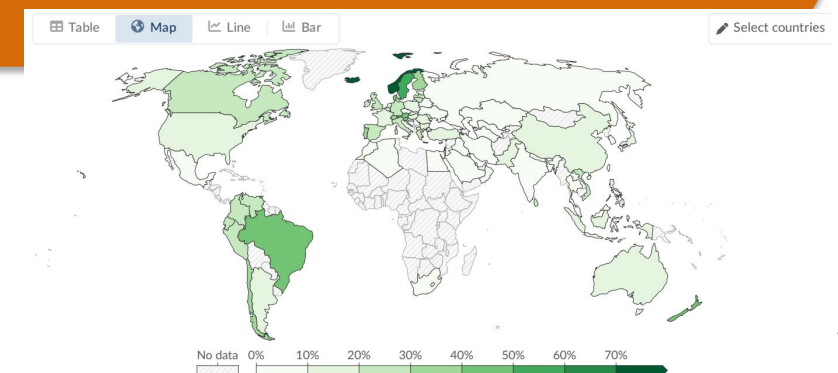
~21% of renewable electricity

Fastest-growing source in history

Driven by falling costs & policy support

Combined, wind and solar nearly match hydropower output — yet growing far faster.

Source: Our World in Data — Wind power generation by country, 2025



Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin Ame

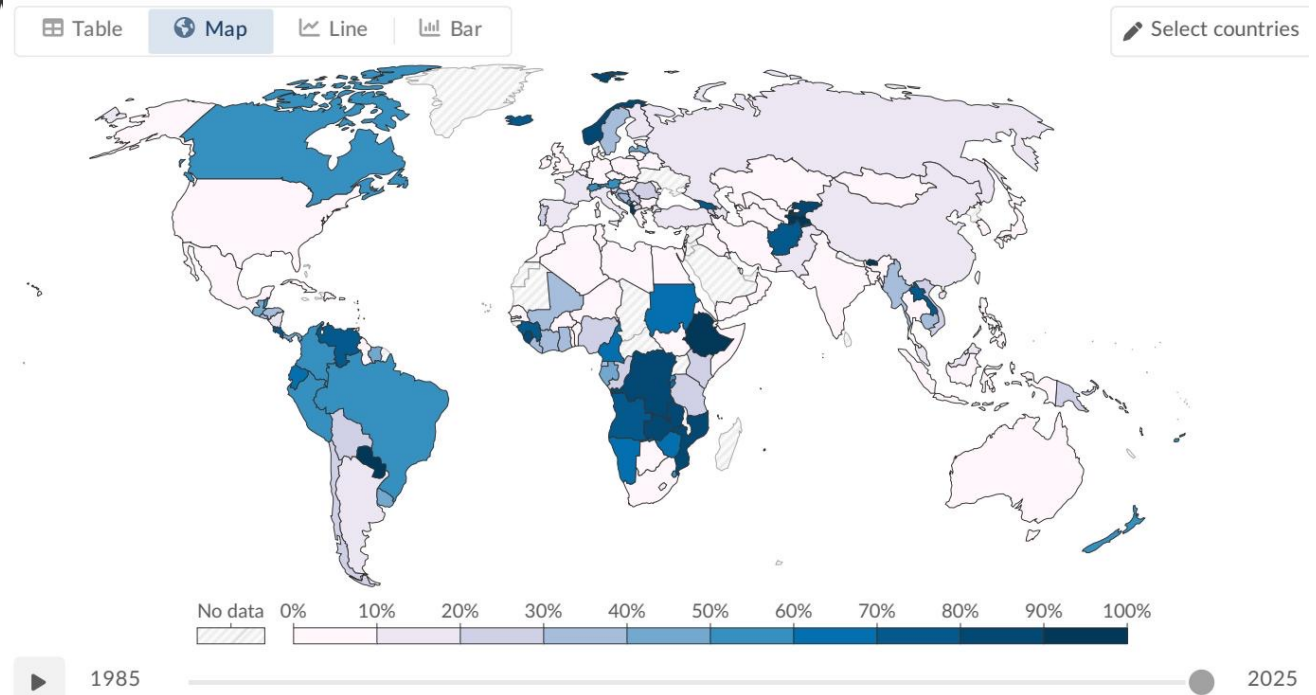
The energy transition is highly uneven — hydropower access and renewable investments vary dramatically by region

Leaders:

- Scandinavia & Canada: Abundant hydropower resources, 60–98% renewable electricity
- Brazil: ~85% renewable electricity from hydropower
- Denmark & Germany: Heavy wind and solar investments — leading the transition in fossil-heavy economies

Laggards:

- Sub-Saharan Africa: Minimal renewable electricity infrastructure, below 10% in many countries
- Middle East: Saudi Arabia, UAE, and others remain almost entirely fossil-fuel dependent (<2%)
- Parts of Asia: Rapid growth but still low overall share in India (~25%), China (~35%)



Source: Our World in Data — Share of electricity from hydropower by country, 2025

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

Five critical findings from the state of global renewable energy in 2025

~14% of primary energy

~30% of electricity

Renewables now supply approximately one-seventh of global primary energy and almost one-third of electricity — a milestone, but far from sufficient for climate targets.

Hydropower dominates

~47% of generation

Hydropower remains the single largest renewable electricity source, accounting for roughly half of all renewable generation. Top producers: China, Brazil, Canada.

Wind + Solar nearly match hydro

Fastest-growing in history

Wind (~2,400 TWh) and solar (~2,000 TWh) combined now nearly match hydro's output and are growing far faster, driven by cost reductions and policy support.

Regional gaps are stark

<2% to ~98%

Scandinavia and Latin America exceed 60–90% renewable electricity while much of the Middle East and Sub-Saharan Africa remain below 10%.

Transport & heating lag

Slowing the full transition

Electricity has advanced most; transport and heating remain heavily reliant on oil and gas, keeping the overall primary energy share far lower.

To meet climate targets, the renewable transition must accelerate dramatically across all sectors — electricity, transport, and heating.